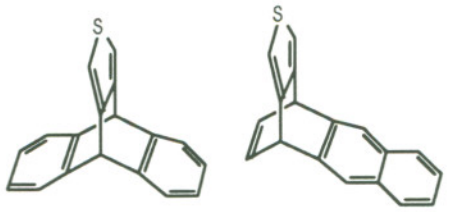
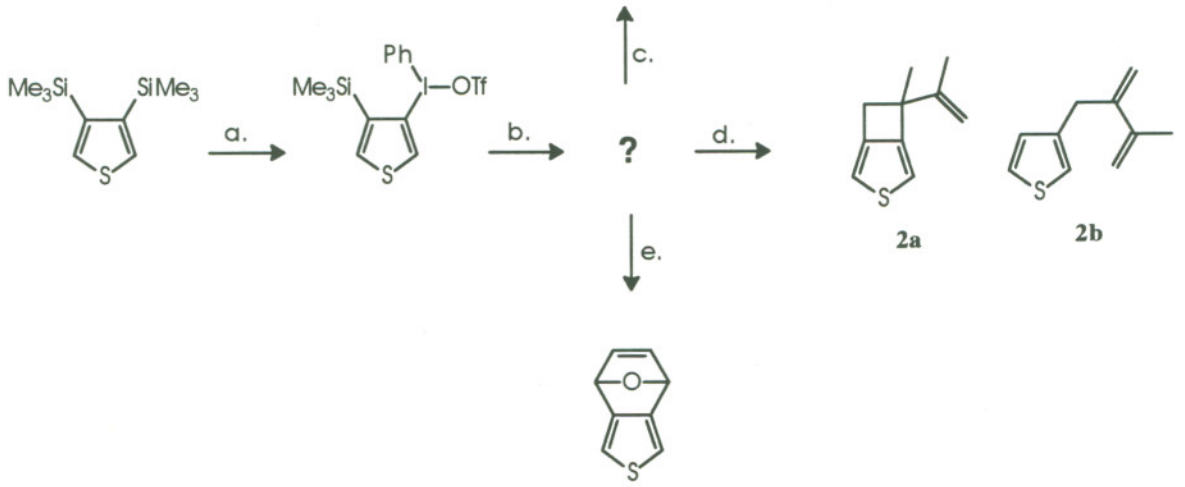
Organic Chemistry Cumulative Examination
March 30, 1996

1. JACS **1996**, *118*, 2511–2512.



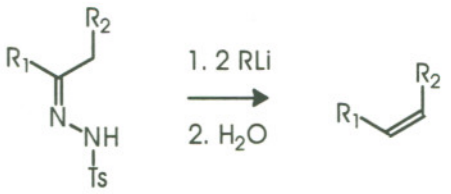
1a **1b**



Reagents/conditions: a. $PhI(OAc)_2$., TfOH, CH_2Cl_2.. b. KF, 18-crown-6, CH_2Cl_2, room temperature. c. Anthracene. d. CH_2=CMeCMe=CH_2. e. Furan

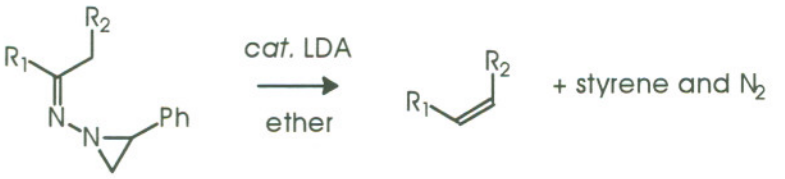
a. Propose a structure for the intermediate represented as "**?**"
b. Explain the formation of each of the products **1a**, **1b**, **2a**, **2b**, and **3** from that intermediate.
c. For the reaction with anthracene, do you expect **1a** or **1b** to be the major product? Why?

2. In our latest organic seminar two weeks ago, we encountered the Shapiro reaction:



Its usefulness is due to the fact that aldehydes or ketones can be easily converted to the tosylhydrazone, so this is a good method of converting carbonyl compounds to olefins. One of its shortcomings is the necessity of using stoichiometric amounts of the alkyllithium reagent.

Recently (JACS **1996**, *118*, 2289–2290) a variant of the Shapiro reaction was described. It uses a phenylaziridinylhydrazone functionality to replase the tosylhydrazone, and uses lithium dialkyl amide (such as LDA) in place of alkyl lithium as the base. The advantage of this procedure is that it only requires catalytic amounts of the base.



a. Write the mechanism of the original Shapiro reaction.
b. Write the mechanism of the catalytic version. Make sure your mechanism explains how it is catalytic.